8. Scenario: You are a data scientist working for a company that sells products online. You have been tasked with analyzing the sales data for the past month. The data is stored in a Pandas data frame. Question: How would you find the top 5 products that have been sold the most in the past month?

AIM:

To identify the top five products with the highest sales quantities during the past month using Pandas.

PROCEDURE:

1. Import the Pandas library.
2. Create a DataFrame containing order dates, product names, and quantities sold.
3. Convert the order date column to datetime format.
4. Filter the records for the required month.
5. Group the data by product and calculate total quantities sold.
6. Sort the products in descending order and display the top five.

PROGRAM CODE:

```
import pandas as pd

sales_data = pd.DataFrame({

   'Order_Date': ['2026-09-05', '2026-09-10',

                  '2026-09-15', '2026-09-20',

                  '2026-09-25', '2026-09-28'],

   'Product': ['Laptop', 'Phone', 'Laptop',
```

```
            'Tablet', 'Phone', 'Headphones'],
    'Quantity': [10, 20, 15, 12, 8, 18]
})
sales_data['Order_Date'] =
pd.to_datetime(sales_data['Order_Date'])
start_date = '2026-09-01'
end_date = '2026-09-30'
monthly_data = sales_data[
    (sales_data['Order_Date'] >= start_date) &
    (sales_data['Order_Date'] <= end_date)
]
top_products = (
    monthly_data.groupby('Product')['Quantity']
    .sum()
    .sort_values(ascending=False)
    .head(5)
)
print("Top 5 Best-Selling Products:")
print(top_products)
```

INPUT:

Product	Quantity sold
Laptop	10
Phone	20
Laptop	15
Tablet	12
Phone	8
Headphones	18

OUTPUT:

```
]

top_products = (
    monthly_data.groupby('Product')['Quantity']
    .sum()
    .sort_values(ascending=False)
    .head(5)
)

print("Top 5 Best-Selling Products:")
print(top_products)
```

```
Top 5 Best-Selling Products:
Product
Phone         28
Laptop        25
Headphones    18
Tablet        12
Name: Quantity, dtype: int64
```

RESULT:

Thus, the best-selling products were identified by calculating total quantities sold and sorting them in descending order. Only four distinct products are present in the sample data, so four products are displayed instead of five.